



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

DDA2001: Introduction to Data Science

Lecture 4: Elementary Probability Theory (continued)

Shuang Li

Recap: 1 - Definition of Probability

What is Probability?

- An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a **random experiment**.
- Probability is used to quantify the likelihood, or chance, that an outcome of a random experiment will occur.

What is Probability?

- Random Experiment:
- Consider one possible outcome: ω
- The outcome ω happens with probability $P(\omega)$
- **It means:**
 - If we repeat such experiment **N** times
 - We observe **n** observations that the outcome is ω .
 - Then if N goes to infinity, **n/N** will approach $P(\omega)$.

Terminologies

- **Random Experiment:** a repeatable procedure
- **Sample space:** a set of all possible outcomes Ω .
- **Event:** a subset of the sample space.
- **Probability function,** $P(\omega)$: gives the probability for each outcome $\omega \in \Omega$
 - Probability is between 0 and 1
 - Total probability of all possible outcomes is 1.
 - If $A = \{\omega_1, \omega_2, \omega_3, \dots\}$, $P(A) = P(\omega_1) + P(\omega_2) + P(\omega_3) + \dots$

Sample space - examples

- Consider the random experiment in which items are selected from a batch consisting of three items {a,b,c}
- Case 1: select two items **without replacement**

Sample space $\{ab, ac, ba, bc, ca, cb\}$

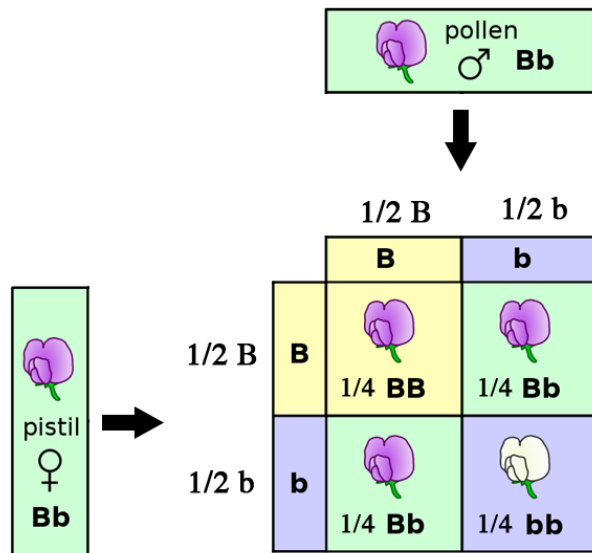
- Case 2: select two items **with replacement**

Sample space $\{aa, ab, ac, ba, bb, bc, ca, cb, cc\}$

Sample space - examples

Recall the heredity example in Lecture 2.

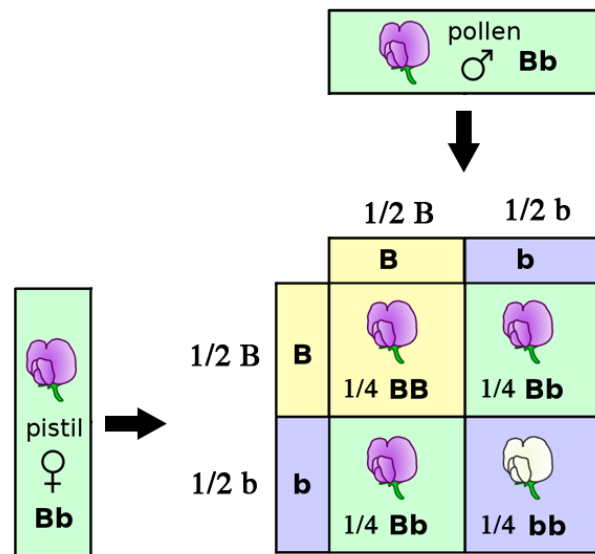
Describe the set of possible outcomes when a pistil of Bb and a pollen of Bb crosses.



Sample space - examples

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Describe the set of possible outcomes when a pistil of Bb and a pollen of Bb crosses.



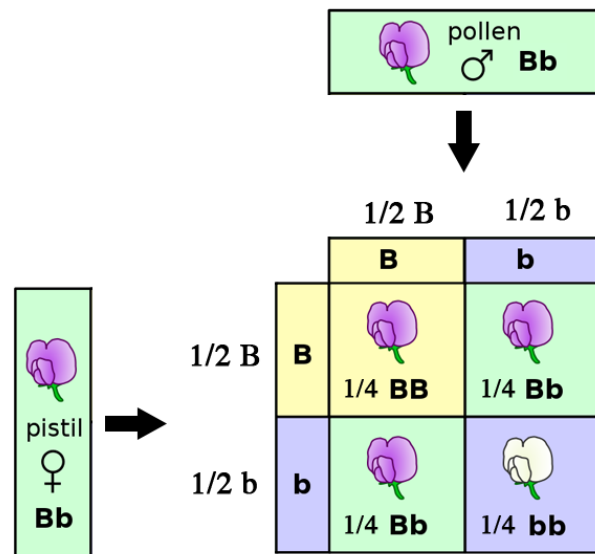
Case 1: Bb and bB are equivalent

Sample space: $\{BB, Bb, bb\}$

Sample space - examples

Recall the heredity example in Lecture 2.

Describe the set of possible outcomes when a pistil of Bb and a pollen of Bb crosses.



Case 2: Bb and bB are not equivalent

Sample space: $\{BB, Bb, bB, bb\}$

Events

- Events are sets:
 - ✓ Can describe in words
 - ✓ Can describe in notation
- Experiment: toss a coin 2 times.
- Event -- You get 1 or more heads
= {HH, HT, TH}

Set operations

- Events are sets, so we can use set operations
 - ✓ Unions
 - ✓ Intersections
 - ✓ Complements

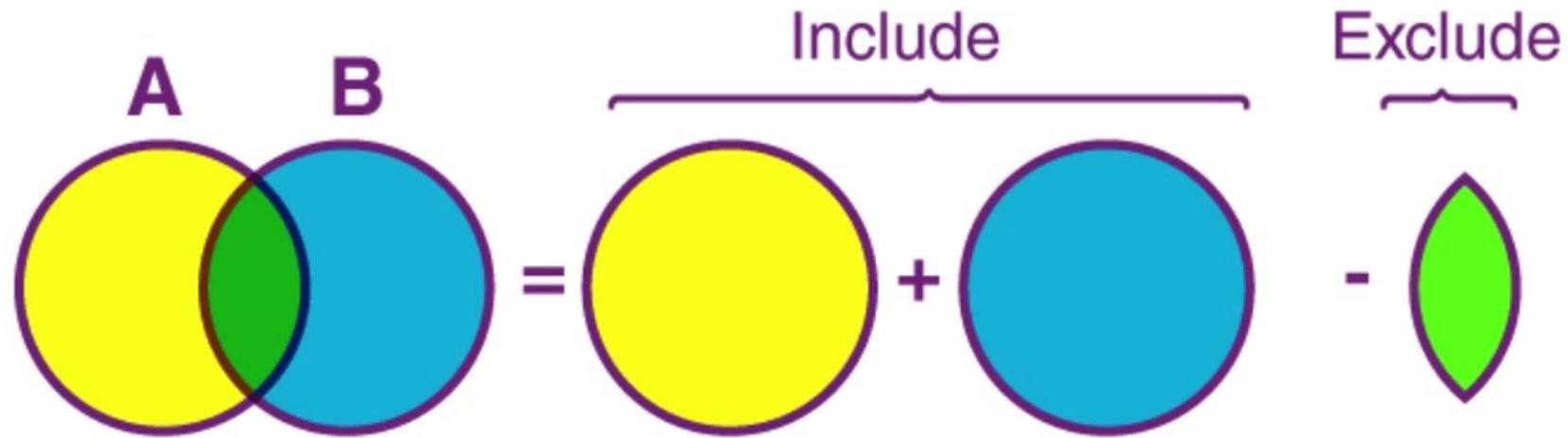
Set operations

- We denote the union as (A or B) in words, and $A \cup B$ in notation
- We denote the intersection as (A and B) in words, and $A \cap B$ in notation
- Two events A and B, such that $A \cap B = \emptyset$ are said to be **mutually exclusive**.
- We denote the complement as (not A) in words, and A' or A^c in notation

Set operations

- The commutative laws: $A \cup B = B \cup A$, $A \cap B = B \cap A$
- The associative laws: $(A \cup B) \cup C = A \cup (B \cup C)$,
 $(A \cap B) \cap C = A \cap (B \cap C)$
- The distributive laws: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$,
 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- De Morgan's laws: $(A \cup B)' = A' \cap B'$, $(A \cap B)' = A' \cup B'$

Set operations

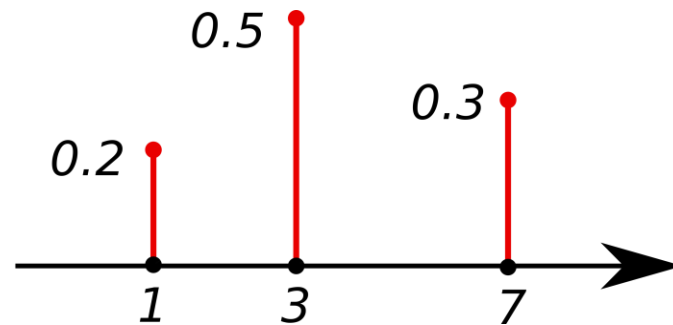


$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Probability function

- Discrete:
 - ✓ **Probability mass function.**
 - ✓ $P(\omega)$: gives the probability for **each** outcome $\omega \in S$



Continuous case will be defined later.

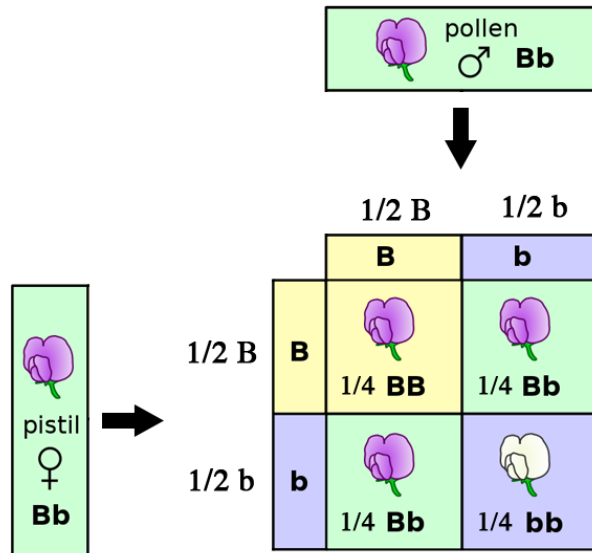
Probability Function- examples

Recall the heredity example in Lecture 2.

Write down $P(\omega)$ for each $\omega \in S$

Case 1: Bb and bB are equivalent

Sample space: $\{BB, Bb, bb\}$

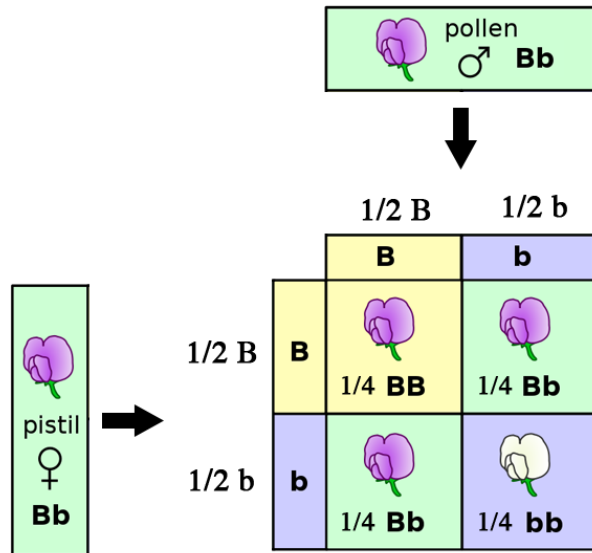


Probability Function- examples

Recall the heredity example in Lecture 2.

Write down $P(\omega)$ for each $\omega \in S$

Case 1: Bb and bB are equivalent



Sample space: $\{BB, Bb, bb\}$

$$P(BB) = 1/4$$

$$P(Bb) = 1/2$$

$$P(bb) = 1/4$$

Today's Topics:

- Independence
- Probability Distributions (CDF)
- Mean and Variance

1. Independence

Independence

- Two events: A and B
- Does knowing something about A tell us whether B happens (and vice versa)?

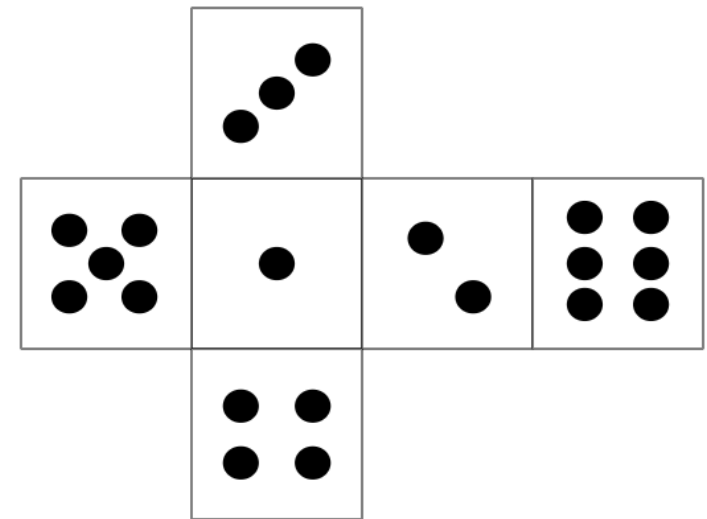
Independence

- Two events: A and B
- Does knowing something about A tell us whether B happens (and vice versa)?

- **Independent**

- A: the first number is 1;
 - B: the second number is 1

$$P(A \text{ and } B) = P(A) \cdot P(B)$$



Two events A, and B are independent if and only if $P(A \cap B) = P(A) \cdot P(B)$

Does knowing something about A tell us whether B happens (and vice versa)?

- **Dependent**

- A: the first number is 1;
- B: the sum of numbers is larger than 2

$$P(A) = 1/6$$

$$P(B) = 35/36$$

$$P(A \text{ and } B) = 5/36$$

$$P(A \text{ and } B) \neq P(A) \cdot P(B)$$

**SAMPLE SPACE FOR
A PAIR OF DICE**

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Comparison: Independence vs Mutually exclusive

- Independence: $P(A \text{ and } B) = P(A)P(B)$
- Mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$

2. (Discrete) Random Variable and Probability Distributions.

In applications we are interested in
quantitative properties of experimental results.

Example 1

- Toss a coin three times and count the number of heads.
- The sample space is

$$S = \{(t, t, t), (t, t, h), (t, h, t), (h, t, t), (t, h, h), (h, t, h), (h, h, t), (h, h, h)\}$$

- Let X count the number of heads. Thus if $s = (t, t, h)$ occurs then $X(s) = 1$.
- X is called a random variable as it takes a numerical value that depends on the outcome of an experiment.

Examples 2:

Roll a fair dice twice, and let the random variable **X** denote the summation of the two numbers.

- What is the **range** (possible values) of the random variable? (think about the sample space)
- How can you describe the event that the summation is larger than 10 using the random variable X?

Examples 2:

Roll a fair dice **twice**, and let the random variable **X** denote the **summation** of the two numbers.

- What is the **range** (possible values) of the random variable? (think about the sample space)
 $\{2,3,4,5,6,7,8,9,10,11,12\}$
- How can you describe the event that the summation is larger than 10 using the random variable X?

Examples 2:

Roll a fair dice twice, and let the random variable **X** denote the summation of the two numbers.

- What is the **range** (possible values) of the random variable? (think about the sample space)

$$\{2,3,4,5,6,7,8,9,10,11,12\}$$

- How can you describe the event that the summation is larger than 10 using the random variable X?

$$\{\omega: X(\omega) > 10\} \text{ or } \{X > 10\}$$

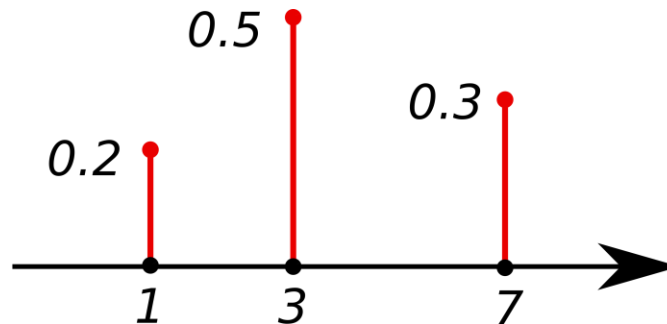
Examples 3:

A group of 10,000 people are tested for a gene called Ifi202 that has been found to increase the risk for lupus. The random variable X is the number of people who carry the gene.

- What is the range (possible values) of the random variable? (think about the sample space)
- How can you describe the event that more than half of the people carry the gene using the random variable X ?

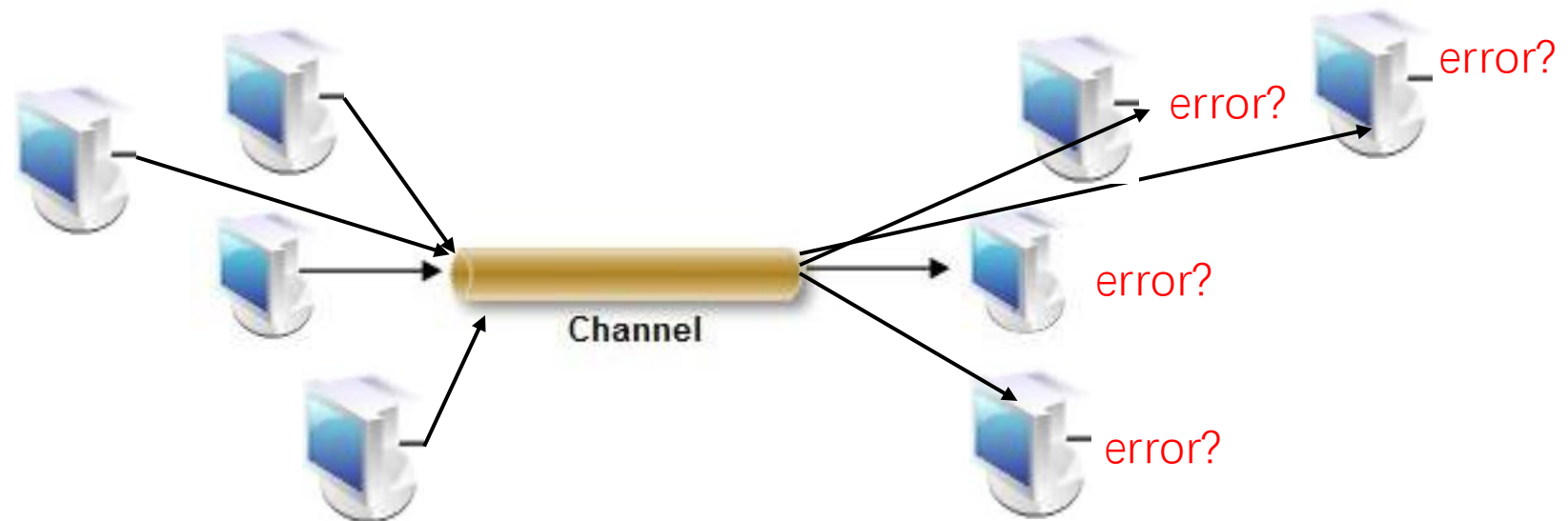
Probability Distributions

- The probability distribution of a random variable X is a description of the probabilities associated with the possible values of X .
- For **discrete** random variables, the distribution is just a list of values, e.g., $\{0.2, 0.5, 0.3\}$. This is called the **probability mass function**.



Probability Distributions - Example

- There is a chance that a bit transmitted through a digital transmission channel is received in error.
- Let X equals the number of bits in error in the next four bits transmitted.
- Q: What is the possible values for X ?

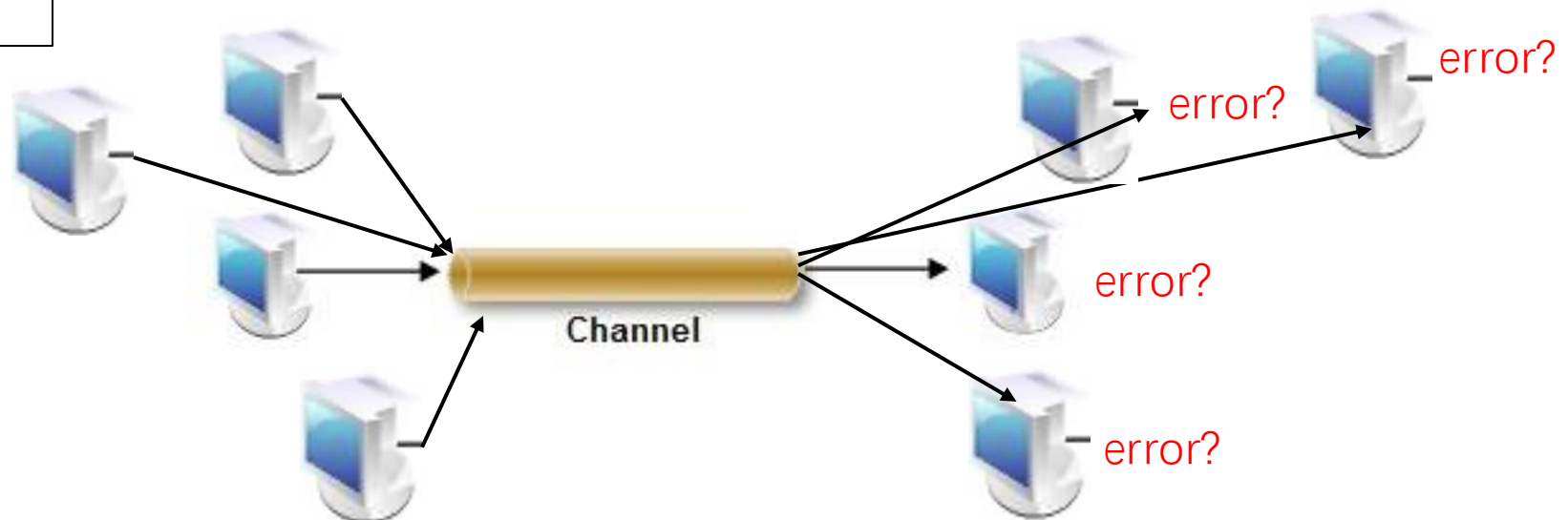


Probability Distributions - Example

- The expert gives the following probability distribution on X .

$$\begin{array}{lll} P(X = 0) = 0.6561 & P(X = 1) = 0.2916 & P(X = 2) = 0.0486 \\ P(X = 3) = 0.0036 & P(X = 4) = 0.0001 & \end{array}$$

Probability that no error received.



Probability Distributions - Example

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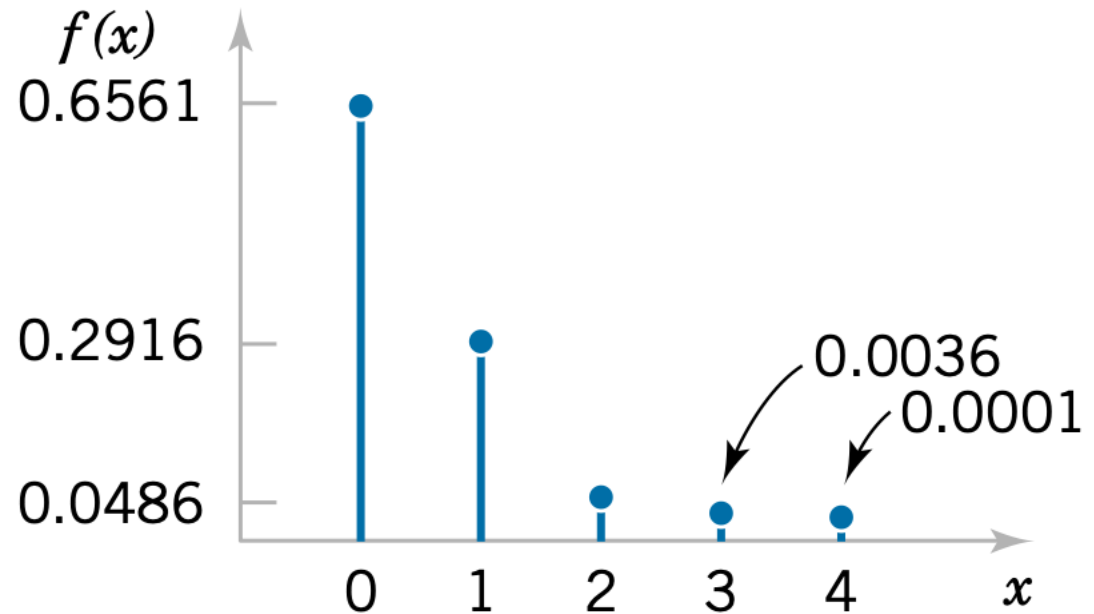
$$P(X = 1) = 0.2916$$

$$P(X = 2) = 0.0486$$

$$P(X = 3) = 0.0036$$

$$P(X = 4) = 0.0001$$

Probability that no error received.



Probability distribution for bits in error

Formal definition

- For a discrete random variable X with possible values $x_1, x_2, \dots, x_n$. A probability mass function $f(\cdot)$ is a function such that:

- ✓ $f(x_i) \geq 0$ for all $x_1, x_2, \dots, x_n$.
- ✓ $\sum_{i=1}^n f(x_i) = 1$
- ✓ $f(x_i) = P(X = x_i)$ for all $x_1, x_2, \dots, x_n$.

Probability that the random variable takes value x_i .

Q: What is the probability mass function f for the previous digital transmission example?

Exercise

- The sample space of a random experiment is $\{a,b,c,d,e,f\}$, and each outcome is **equally likely**. A random variable X is defined as follows (think X is the pay you get by observing each outcome):

outcome	a	b	c	d	e	f
x	0	0	1.5	1.5	2	3

- Q: What is the probability mass function f of X ?
- Use the probability mass function to determine:
 - (a) $P(X = 1.5)$
 - (b) $P(0.5 < X < 2.7)$
 - (c) $P(X > 3)$
 - (d) $P(0 \leq X < 2)$
 - (e) $P(X = 0 \text{ or } X = 2)$

Exercise

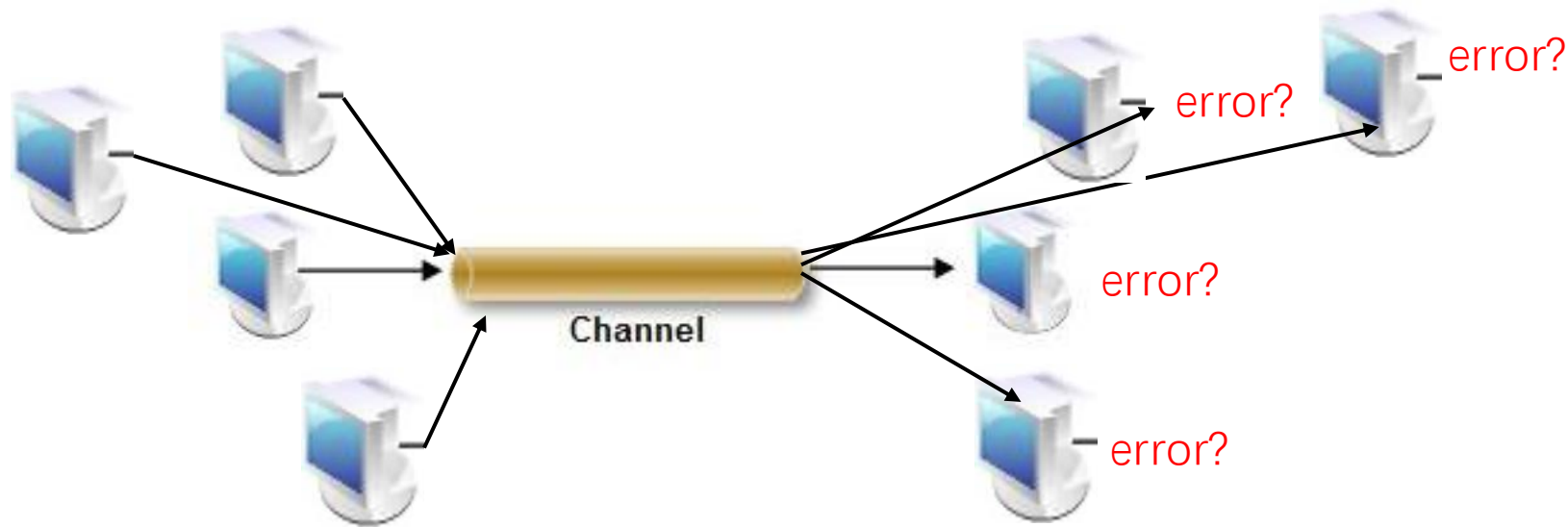
- Verify that the following function f is a probability mass function, and determine the requested probabilities.

$$f(x) = \frac{2x + 1}{25}, \quad x = 0, 1, 2, 3, 4$$

- | | |
|-----------------------|-------------------|
| (a) $P(X = 4)$ | (b) $P(X \leq 1)$ |
| (c) $P(2 \leq X < 4)$ | (d) $P(X > -10)$ |

Cumulative Distribution Functions (CDF)

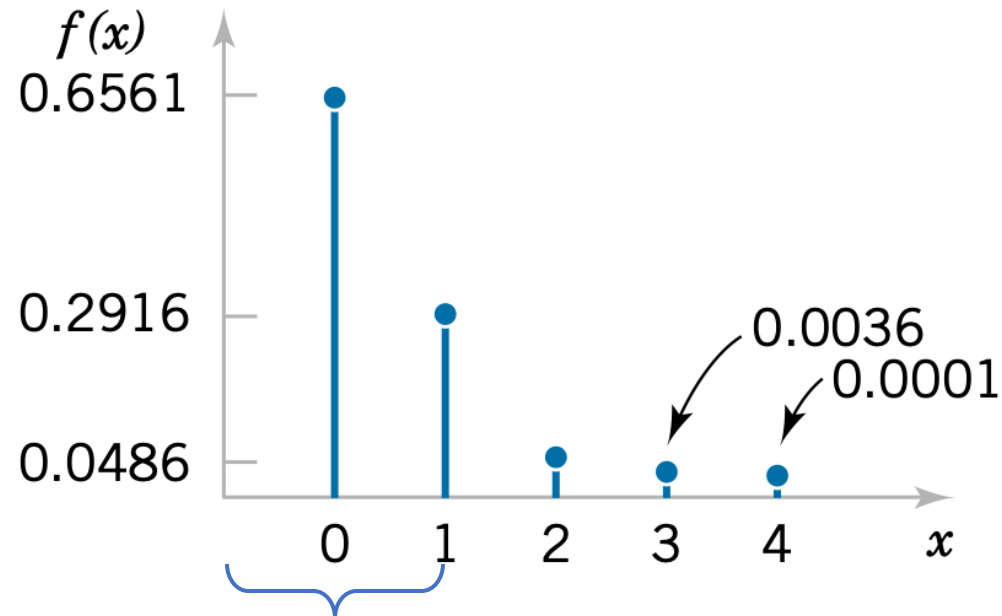
- Recall the digital transmission example.



- What is the probability of having three or fewer bits in error?
- What about two or fewer?
- What about four or fewer? ➡ Cumulative probabilities

Cumulative Distribution Functions (CDF)

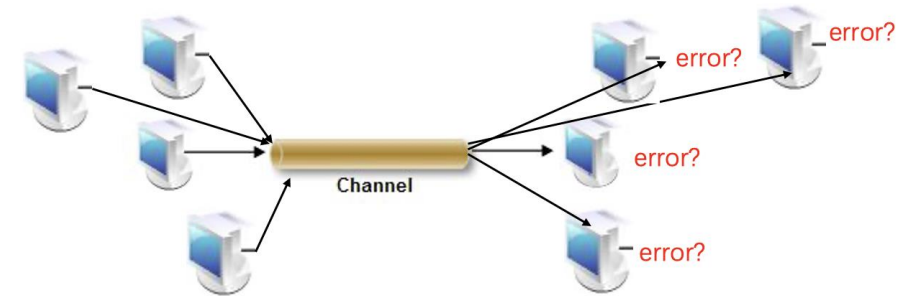
- Cumulative Probabilities



probability of having one or fewer bits in error: $0.6561 + 0.2916$

probability of having two or fewer bits in error: $0.6561 + 0.2916 + 0.0486$

Cumulation!



Cumulative Distribution Functions (CDF)

- The cumulative distribution function (CDF) of a discrete random variable X , denoted as $F(x)$ is

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

- $F(x)$ satisfies:
 - ✓ $0 \leq F(x) \leq 1$
 - ✓ If $x \leq y$, then $F(x) \leq F(y)$

Note: even if the random variable X can only take integer values, the cumulative distribution function can be defined at **non-integer** values, for example, $F(1.5) = P(X = 0) + P(X = 1) = 0.6561 + 0.2916$.

Cumulative Distribution Functions (CDF)

- The probability mass function provides probabilities.
- The cumulative distribution function can also provide probabilities, and it uniquely determines the probability mass function of a discrete random variable.

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

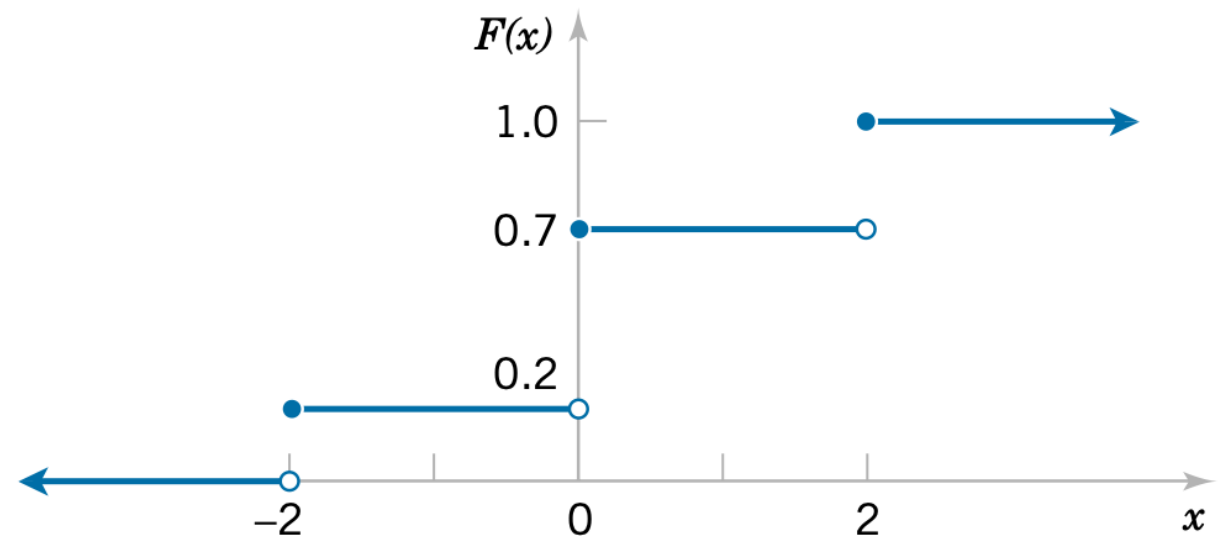
- Why?

Example

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

- Find the probability mass function of X from the following cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -2 \\ 0.2 & -2 \leq x < 0 \\ 0.7 & 0 \leq x < 2 \\ 1 & 2 \leq x \end{cases}$$



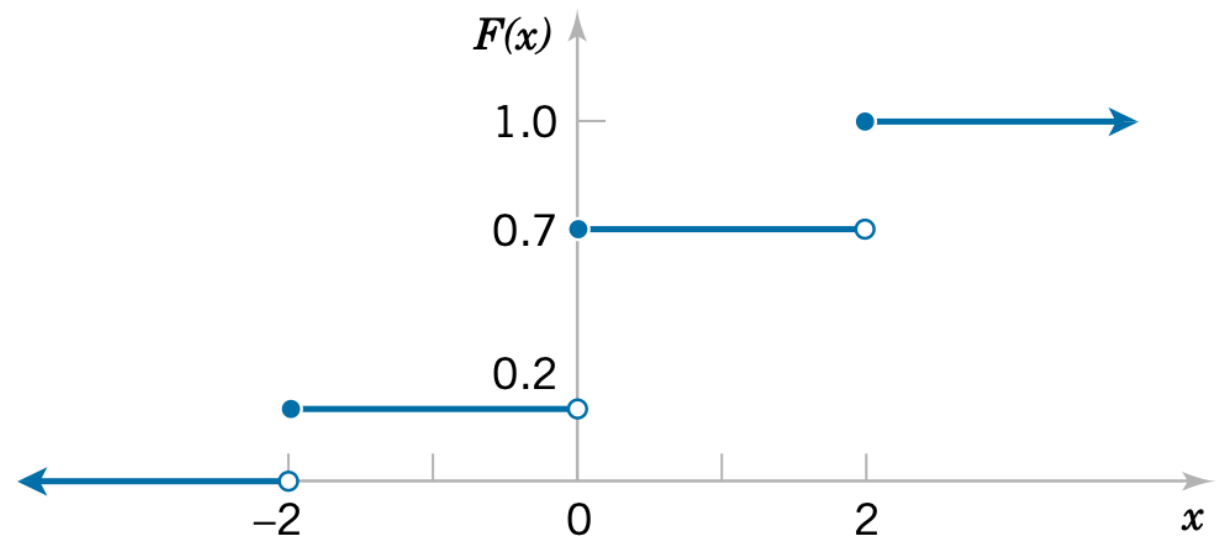
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$$f(-2)=0.2 \quad f(0)=0.5 \quad f(2)=0.3$$



You need to know:

- Given the probability mass function, how do we obtain CDF?
- Given the CDF, how do we obtain the probability mass function?

3. Mean and Variance.

QUESTION

- Which game would you rather play? We flip a fair coin.

Game 1:

- If heads, You pay me \$1.
- If Tails, I pay you \$1.

Game 2:

- If Heads, you pay me \$1000.
- If Tails, I pay you \$1000.1.



QUESTION

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Return
Game 1 < Game 2

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Risk

Game 1 < Game 2

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Return

Game 1 < Game 2

Risk

Game 1 < Game 2

How can we quantify the return and risk?

Return (Expected Value)

The expected return (for you) for both games is calculated as:

$$\text{Expected Return} = \frac{1}{2} \times (\text{Payoff if Heads}) + \frac{1}{2} \times (\text{Payoff if Tails})$$

Game 1:

$$\text{Expected Return} = \frac{1}{2} \times (-1) + \frac{1}{2} \times (1) = 0$$

Game 2:

$$\text{Expected Return} = \frac{1}{2} \times (-1000) + \frac{1}{2} \times (1000.1) = 0.05$$

Return: Game 1 < Game 2

Risk (Variance)

Variance measures the spread of outcomes. The formula is:

$$\text{Variance} = \frac{1}{2} \times (\text{Payoff if Heads} - \text{Expected Return})^2 + \frac{1}{2} \times (\text{Payoff if Tails} - \text{Expected Return})^2$$

Please compute by yourself.

Risk: Game 1 < Game 2

The risk of Game 1 is significantly lower!!

Example: Investment

Maximize average return?

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Maximize average return?

No! Also reduce the chance of losing too much after one investment.

Example: Investment

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No! Also reduce the chance of losing too much after one investment.



Risk

Example: Investment

Maximize average return?

No! Also reduce the chance of losing too much after one investment.



Real objective:

Example: Investment

Maximize average return?

No! Also reduce the chance of losing too much after one investment.



Real objective:

Maximize: Average Return - Risk

Example: Investment

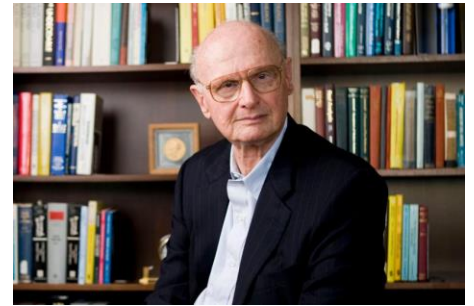
Maximize average return?

No! Also reduce the chance of losing too much after one investment.



Real objective:

Maximize: Average Return - Risk



Markowitz Model

The first model you need to learn to work in finance.

How to calculate Return and Risk?

Game 1:

- If heads, You pay me \$1.
- If Tails, I pay you \$1.
- Return
- Risk

How to calculate Return and Risk?

Game 1:

- If heads, You pay me \$1.
- If Tails, I pay you \$1.

- Return -> Mean

$$E[X] = \sum_x x P(X = x) = \sum_x x f(x)$$

- Risk

How to calculate Return and Risk?

Game 1:

- If heads, You pay me \$1.
- If Tails, I pay you \$1.

- Return -> Mean

$$E[X] = \sum_x x P(X = x) = \sum_x x f(x) = 1 \times \frac{1}{2} + (-1) \times \frac{1}{2} = 0$$

- Risk

How to calculate Return and Risk?

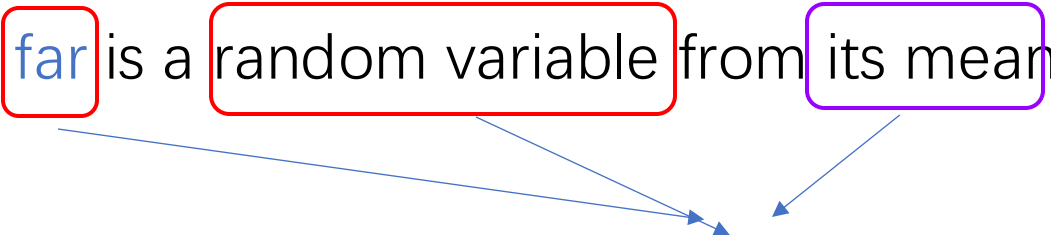
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- Return -> Mean

$$E[X] = \sum_x x P(X = x) = \sum_x x f(x) = 1 \times \frac{1}{2} + (-1) \times \frac{1}{2} = 0$$

- Risk -> How far is a random variable from its mean, on average?


$$|X - E[X]|$$

How to calculate Return and Risk?

Game 1:

- If heads, You pay me \$1.
- If Tails, I pay you \$1.

- Return -> Mean

$$E[X] = \sum_x x P(X = x) = \sum_x x f(x) = 1 \times \frac{1}{2} + (-1) \times \frac{1}{2} = 0$$

- Risk -> Variance

$$\text{Var}[X] = \sum_x (x - E[X])^2 f(x)$$

How to calculate Return and Risk?

Game 1:

- If heads, You pay me \$1.
- If Tails, I pay you \$1.

- Return -> Mean

$$E[X] = \sum_x x P(X = x) = \sum_x x f(x) = 1 \times \frac{1}{2} + (-1) \times \frac{1}{2} = 0$$

- Risk -> Variance

$$\text{Var}[X] = \sum_x (x - E[X])^2 f(x) = 1^2 \times \frac{1}{2} + (-1)^2 \times \frac{1}{2} = 1$$

Game 2:

- If heads, You pay me \$1000.
- If Tails, I pay you \$1000.1.

Compute the mean and variance by yourself.

Summary: Mean and Variance

- Mean

$$E[X] = \sum_x x P(X = x) = \sum_x x f(x)$$

- Variance

$$\text{Var}[X] = \sum_x (x - E[X])^2 f(x)$$